Agentic Predictor

Multi-View Encoders for Performance Prediction in LLM-Based Agentic Workflows

Patara Trirat[1] Wonyong Jeong[1] Sung Ju Hwang[1 2]

[1] DeepAuto.ai
[2] KAIST

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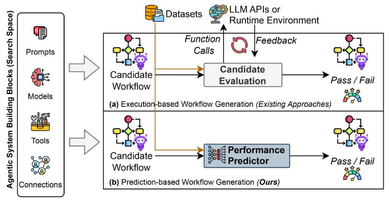
Motivation

LLM-powered agentic systems require complex workflow design.
Current optimization methods rely on costly execution-based evaluations.

- Vast configuration space (prompts, agents, tools, etc.)
- High cost of runtime evaluations for workflow selection
- Need for task-specific adaptation

Our Goal

Enable efficient agentic workflow *search and evaluation* via learned prediction models.



Key Contributions

We propose **Agentic Predictor**, a lightweight, predictive framework to estimate the success of agentic workflows using *multi-view representation* learning and unsupervised pretraining.

✅ **Multi-view encoding** captures workflow heterogeneity from:
- Graph structure (agent interaction)
- Code semantics (logic & tool use)
- Prompt embeddings (roles & behaviors)

✅ **Cross-domain unsupervised pretraining** trains encoder on unlabeled workflows from various domains.
✅ **Lightweight performance predictor** guides search efficiently using minimal labeled data.

Table 2: Summary of benchmark statistics.

Domains	Code Generation (GD/AF)	Math (GD/AF)	Reasoning (GD/AF)
# workflows	739 / 38	300 / 42	189 / 30
Avg. # nodes	5.96 / 6.11	6.06 / 5.49	5.97 / 6.58
# tasks	233 / 233	782 / 782	2,400 / 2,400
# samples	30,683 / 7,362	12,561 / 4,059	453,600 / 72,000

HumanEval MBPP GSM8K MATH MMLU

Evaluation Metrics

Accuracy quantifies how well a model predicts agentic workflow performance.
Utility evaluates the consistency between the workflow rankings predicted by the model and the ground-truth rankings, emphasizing the model's ability to determine the relative order of different workflows.

Ablation Study

Table 4: Results of ablation study on different input view variations.

Variations			Code Generation		Math Problem		Common Reasoning		Average	
Code	Graph	Text	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility
✓			82.04	75.66	75.70	68.52	83.19	**91.51**	80.31	78.56
	✓		84.44	77.22	79.14	67.99	87.00	91.03	83.53	78.75
		✓	79.87	70.34	76.60	68.45	68.06	71.04	74.84	69.94
✓	✓		83.72	73.97	75.86	70.18	86.88	86.14	82.15	76.76
✓		✓	82.27	77.28	76.03	66.66	54.17	53.21	70.82	65.72
	✓	✓	82.45	74.64	75.70	67.83	69.47	70.55	75.87	71.01
✓	✓	✓	**85.62**	**80.08**	**79.56**	**74.08**	**87.96**	91.47	**84.38**	**81.88**

Table 5: Results of ablation study on different input graph variations.

Variations		Code Generation		Math Problem		Commong Reasoning		Average	
Single View	Multi View	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility
✓		82.58	**78.52**	78.57	67.51	86.95	90.14	82.70	78.72
	✓	**84.44**	77.22	**79.14**	**67.99**	**87.00**	**91.03**	**83.53**	**78.75**

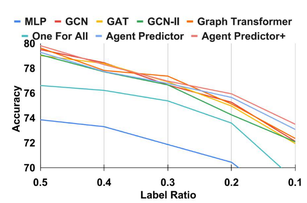


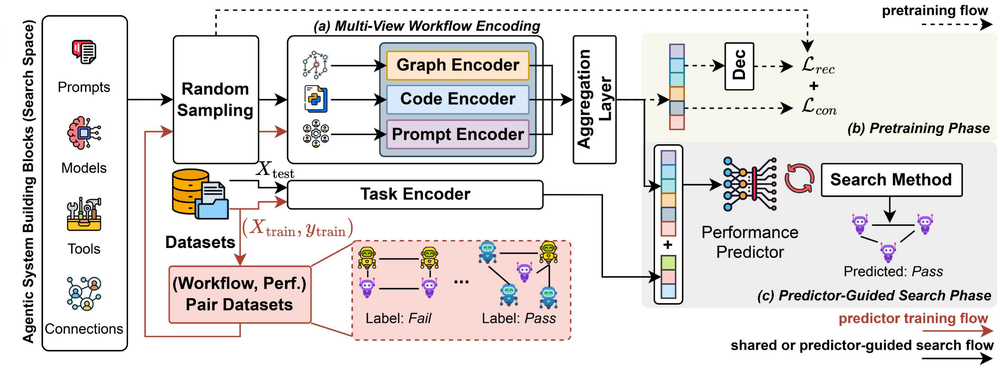
Figure 3: Accuracy comparison between Agentic Predictor and the baselines across varying label ratios.

Framework Overview

Project Page

Encoder	Responsibility
Graph Encoder	Explicitly models structural dependencies and direct interactions among agents, emphasizing inter-agent communication channels.
Code Encoder	Implicitly encodes program-level semantics, control and logical sequence, computational complexity, and patterns of tool usage inherent in workflow implementations.
Prompt Encoder	Provides semantic embeddings that capture agent roles, behavioral specifications, and broader contextual guidance embedded within system and instruction prompts.

➕ **CrossGraphAttn** enriches each node with multi-view contextual dependencies (attending across views for each node).
➕ **ViewAttnPool** highlights which views are most informative.
➕ **Aggregation Layer** integrates information across all views, enabling the model to emphasize the most contextually relevant aspects.
➕ **Cross-Domain Multi-Task Pretraining** encourages the encoder to capture meaningful signals related to both structure and performance.



Overall Procedures: A (a) multi-view workflow encoder is designed to encode a set of agentic workflows from graph, code, and prompt aspects into unified representations, which serve as features for training the predictor. In the (b) pretraining phase, the encoder learns these representations on unlabeled workflows spanning diverse tasks and domains, using cross-domain unsupervised pretraining objectives. In the (c) predictor-guided search phase, a performance predictor is trained on a small (workflow configuration, performance) dataset to classify configurations as pass or fail, and subsequently guides the search toward promising configurations.

Experimental Results

Table 3: Performance comparison between Agentic Predictor and baselines. The best and second-best results are highlighted in **bold** and underlined, respectively. GD is G-Designer, and AF is AFlow.

Domain	CodeGD		CodeAF		MathGD		MathAF		ReasonGD		ReasonAF		Average	
Model	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility
MLP	83.88 (±0.04)	76.16 (±0.03)	78.02 (±0.59)	73.94 (±1.35)	63.22 (±0.30)	64.13 (±0.44)	73.73 (±0.31)	69.64 (±0.29)	71.54 (±0.09)	62.41 (±1.67)	78.45 (±0.08)	88.48 (±0.63)	74.81 (±0.24)	72.46 (±0.74)
GCN	84.23 (±0.04)	79.31 (±0.10)	84.35 (±0.34)	72.73 (±1.18)	64.12 (±0.17)	63.03 (±0.59)	76.19 (±0.42)	66.52 (±1.66)	72.22 (±0.03)	59.18 (±0.85)	87.12 (±0.14)	**91.82** (±0.46)	78.04 (±0.19)	72.10 (±0.81)
GAT	85.14 (±0.25)	79.50 (±0.14)	84.49 (±0.56)	76.46 (±0.91)	64.84 (±0.96)	62.32 (±0.93)	76.44 (±0.61)	66.51 (±1.28)	72.16 (±0.03)	59.44 (±1.06)	87.07 (±0.08)	89.40 (±0.68)	78.36 (±0.42)	72.27 (±0.83)
GCN-II	83.81 (±0.07)	78.45 (±0.74)	83.72 (±0.40)	77.75 (±0.98)	63.56 (±0.74)	66.02 (±0.10)	75.04 (±0.31)	64.33 (±0.47)	72.29 (±0.09)	59.10 (±1.02)	87.28 (±0.14)	89.92 (±1.90)	77.62 (±0.29)	72.60 (±0.87)
Graph Transformer	85.24 (±0.19)	80.20 (±0.64)	84.71 (±0.45)	74.09 (±0.35)	63.25 (±0.70)	64.97 (±0.36)	75.45 (±0.23)	66.48 (±0.96)	72.26 (±0.08)	60.92 (±1.79)	86.93 (±0.27)	90.60 (±1.97)	77.97 (±0.32)	72.88 (±1.01)
Dir-GNN	84.85 (±0.11)	79.81 (±0.69)	83.45 (±0.41)	76.08 (±0.92)	63.01 (±0.54)	64.68 (±1.66)	76.11 (±0.65)	67.97 (±0.16)	74.25 (±0.12)	62.64 (±0.92)	86.66 (±0.13)	90.07 (±1.68)	78.05 (±0.33)	73.54 (±1.01)
One For All	83.74 (±0.09)	75.93 (±0.12)	81.05 (±0.34)	73.42 (±1.39)	63.17 (±0.21)	66.65 (±0.82)	75.21 (±0.23)	69.08 (±0.64)	72.29 (±0.12)	60.35 (±1.25)	82.52 (±0.13)	87.64 (±1.98)	76.33 (±0.19)	72.18 (±1.03)
Agentic Predictor	**85.33** (±0.05)	**81.42** (±0.26)	**85.62** (±0.47)	**80.08** (±0.46)	**66.20** (±0.17)	**67.88** (±0.21)	**79.56** (±0.25)	**74.08** (±0.47)	**75.13** (±0.01)	**63.06** (±0.45)	**87.96** (±0.02)	91.47 (±0.44)	**79.97** (±0.16)	**76.33** (±0.38)
Δ vs. best baseline (% Improvement)	+0.09 (0.11%)	+1.22 (1.52%)	+0.91 (1.07%)	+2.33 (3.00%)	+1.36 (2.09%)	+1.23 (1.85%)	+3.12 (4.08%)	+4.44 (6.38%)	+0.88 (1.19%)	+0.42 (0.67%)	+0.68 (0.78%)	-0.35 (-0.38%)	+1.61 (2.05%)	+2.79 (3.79%)

GD and **AF** denote the *independently* developed **G-Designer** (ICML'25) and **AFlow** (ICLR'25) agentic frameworks.

Table 9: Comparison between Agentic Predictor and LLM-based few-show classification.

Domain	Code Generation		Math Problem		Common Reasoning		Average	
Model	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility	Accuracy	Utility
GPT-4.1 (∼$59)	62.42	57.00	67.08	52.97	59.10	66.79	62.86	58.92
Claude 4 Sonnet (∼$202)	56.72	51.65	64.62	57.32	44.50	41.25	55.28	50.07
Gemini 2.5 Flash (∼$21)	60.52	58.94	51.60	55.21	59.20	63.17	57.10	59.11
Agentic Predictor	**84.40**	**78.84**	**80.10**	**77.61**	**90.40**	**87.67**	**84.97**	**81.37**

Computation Cost

Model	Training		Inference	
	Time (s/epoch)	Memory (GB)	Time (ms/sample)	Memory (GB)
MLP	0.195	0.033	0.002	0.020
GCN	4.867	0.058	0.017	0.040
GAT	5.108	0.058	0.023	0.042
GCN-II	4.623	0.058	0.015	0.040
Graph Transformer	5.372	0.087	0.023	0.060
Dir-GNN	4.965	0.077	0.023	0.050
One For All	6.140	0.038	0.018	0.038
GPT-4.1 Claude 4 Sonnet Gemini 2.5 Flash	N/A (via OpenRouter API)		2253.333 1888.333 2606.667	N/A
Agentic Predictor (+ pretraining)	4.840 (168.104)	2.760 (13.520)	0.054	0.490

Result Summary

- State-of-the-art performance across Code, Math, and Reasoning benchmarks.
- +1.6% accuracy / +2.8% utility over strongest graph baselines.
- Multi-view encoding (graph + code + prompt) is essential for predicting agentic workflow performance.
- Cross-domain unsupervised pretraining enables strong performance under limited labels.
- Structured predictors outperform few-shot LLM scoring, while being orders of magnitude cheaper.